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The Behavioral Sufficiency Problem

Why AI Governance Frameworks, Modeled on Human Regulatory Theory, Cannot Operate Without the Cultural and Social Infrastructure That Co-Evolved Alongside Human Law

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Data Sources

Oxford Insights Government AI Readiness Index 2024 (188 countries) + AI Incident Database (1,362 incidents through February 14, 2026) + AIID Report Metadata (7,967 records)

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AI governance; behavioral economics; AI safety; incident documentation; reporting bias; AI policy; regulatory theory; organizational culture

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ABSTRACT

AI governance frameworks have become the dominant policy response to AI risk across government, defense, and commercial sectors. This paper argues that these frameworks suffer from a fundamental structural problem: they are modeled on human regulatory theory, which was never designed to operate without the cultural, social, and psychological infrastructure that co-evolved alongside human law. We term this the Behavioral Sufficiency Problem.

Using original empirical analysis merging the Oxford Insights Government AI Readiness Index 2024 with the AI Incident Database (1,362 incidents through February 2026), we find a statistically significant positive correlation between governance maturity and documented public safety AI incidents ($r = 0.363$, $p = 0.021$). This counterintuitive result reveals that governance primarily creates reporting infrastructure rather than preventing incidents.

We present two analytical lenses through which to understand the Behavioral Sufficiency Problem and show that both converge on the same conclusion: current governance frameworks are not sufficient to determine AI system behavior in high-stakes domains. We conclude with practical prescriptions for policymakers, deploying organizations, and AI developers.

1. INTRODUCTION

The past five years have produced an extraordinary proliferation of AI governance frameworks. From the OECD AI Principles (2019) to the EU AI Act (2024), from the US Executive Order on AI (2023) to hundreds of national strategies and sector-specific guidelines, the dominant policy response to AI risk has been the construction of frameworks, principles, regulations, and standards. The implicit model is clear: if AI systems operate within well-designed governance structures, harmful outcomes will be constrained.

This paper questions that implicit model — not by arguing that governance is irrelevant, but by arguing that it is insufficient. Specifically, we observe that human regulatory theory, from which AI governance frameworks are almost entirely derived, was never designed to operate as the sole behavioral constraint on intelligent agents. Human law is the backstop of a much richer system of behavioral regulation that includes internalized values, professional identity, peer accountability, social shame, community belonging, and moral intuition. Remove those non-governance constraints, and no volume of regulation reliably produces safe behavior.

For AI systems, we are attempting to run the backstop as the only mechanism. We term this the Behavioral Sufficiency Problem.

The problem is not merely theoretical. As we show in Section 3, empirical analysis of cross-national AI incident data reveals that the relationship between governance maturity and documented AI harms is positive, not negative — a finding that is only explicable if governance primarily shapes the visibility of harm rather than its occurrence. This suggests that the current governance-centric approach to AI safety may be producing a dangerous illusion of control: systems that appear well-governed because harms are undetectable, not because harms are absent.

This paper does not argue against AI governance. It argues that governance, as currently conceived and deployed, is a necessary but insufficient condition for AI safety in high-stakes domains. Understanding what the sufficient conditions are — and what the human analogs look like — is the central task of this research program.

2. BACKGROUND AND RESEARCH QUESTIONS

2.1 The Governance-Centric Paradigm

Contemporary AI policy has converged on what we term the governance-centric paradigm: the view that AI risk is primarily a problem of regulatory design. This paradigm is evidenced by the structure of major AI policy initiatives globally. The EU AI Act establishes a risk-tiered compliance framework. The OECD AI Principles articulate normative standards for responsible AI. National AI strategies uniformly include governance chapters addressing transparency, accountability, and human oversight.

The governance-centric paradigm draws implicitly on a well-established tradition in regulatory theory. From Selznick's foundational work on regulatory agencies to modern responsive regulation theory (Ayres & Braithwaite, 1992), regulatory scholarship has developed sophisticated frameworks for designing rules that shape organizational and individual behavior. AI governance inherits this intellectual tradition.

What AI governance frameworks do not adequately inherit is regulatory theory's implicit assumption about behavioral context. Regulatory theory was developed to govern human actors operating within social systems. Human actors are embedded in communities with norms, in professions with cultures, in relationships with accountability, and in identities with moral stakes. Regulation works partly because it operates within this broader behavioral ecosystem. AI systems, as currently deployed, have no analogous ecosystem.

2.2 The Human Behavioral System

Understanding the Behavioral Sufficiency Problem requires mapping the full range of mechanisms that constrain human behavior. Law and formal governance are visible and extensively studied. But they operate within a system of behavioral constraints that includes:

- Internalized values and moral identity — individuals develop stable dispositions toward certain behaviors based on who they understand themselves to be. A physician who would not harm a patient is not primarily constrained by malpractice law; the constraint is constitutive of professional identity.
- Professional ethics culture — occupational communities establish norms, conventions, and standards that operate independently of formal regulation. Medical ethics, legal ethics, and engineering ethics all function as behavioral constraints that are enforced through professional socialization, not only through regulatory mechanisms.
- Peer accountability and social shame — humans are acutely sensitive to the judgments of respected peers. The prospect of professional disapproval, reputation damage, or social exclusion constrains behavior in ways that formal sanctions alone cannot replicate.
- Community feedback loops — social systems provide real-time correction when behavior deviates from norms. Informal feedback, reputation systems, and community response all operate as dynamic behavioral correction mechanisms.
- Narrative and meaning — humans embed their choices in stories about who they are and what they stand for. These narratives create accountability to self that operates independently of external observation or sanction.

Governance frameworks — laws, regulations, guidelines — interact with all of these mechanisms. In well-functioning social systems, formal governance and informal norms are mutually reinforcing. When formal governance is weakened, informal norms provide backup. When informal norms erode, formal governance becomes more necessary. The two systems are not independent.

2.3 Research Questions

This paper addresses three related research questions:

- Empirically: Is there a detectable relationship between national AI governance maturity and documented public safety AI incidents, and if so, what does its direction and magnitude suggest about the mechanism?
- Theoretically: In what ways are AI systems relevantly similar to or different from human agents with respect to behavioral constraint, and what are the governance implications of each framing?
- Prescriptively: If governance alone is insufficient, what constitutes the full set of sufficient conditions for safe AI behavior, and what are the policy and organizational implications?

2.4 Related Work

This paper builds on and extends several lines of prior research that have examined the relationship between AI governance frameworks and real-world AI behavior.

Raji et al. (2025) demonstrated the viability of using the AI Incident Database for governance-relevant analysis, establishing AIID as a credible empirical resource for AI policy research. The present paper adopts and extends this approach by merging AIID incident data with national governance readiness scores to test whether governance maturity predicts incident outcomes—a cross-national empirical test that, to our knowledge, has not been previously attempted.

The AIGN Governance Culture Framework (Upmann et al., 2024) introduced the concept of AI governance culture as a determinant of organizational AI behavior, arguing that compliance alone is insufficient without an enabling culture of responsible AI practice. This insight informs the present paper’s theoretical argument that behavioral norms—not just regulatory frameworks—shape real-world AI safety outcomes. Where the AIGN framework proposes a practitioner model for cultivating governance culture, this paper tests the behavioral hypothesis empirically across national governance systems.

Additionally, the Policy and Society special issue on generative AI governance (Taeihagh, 2025) has examined structural gaps between policy intent and deployment reality, noting the inadequacy of voluntary self-regulatory efforts and proposing adaptive, anticipatory approaches to AI governance. The sufficiency gap identified in the present paper parallels these concerns but arrives at the diagnosis through a different route: rather than proposing a new governance model, we test the existing governance-centric paradigm against empirical incident data and find it wanting.

3. DATA AND METHODOLOGY

3.1 Data Sources

The primary analytical dataset was constructed by merging two public data sources:

Dataset	Coverage	Key Variables
Oxford Insights GARI 2024	188 countries, scored 0–100 on AI readiness dimensions including governance	Composite readiness score, rank, regional classification, governance sub-index
AI Incident Database (AIID)	1,362 incidents from 1983 through February 14, 2026, maintained by the Partnership on AI	Deployer identity, harm type, incident date, linked media reports
AIID Reports Metadata	7,967 individual media report records linked to AIID incidents	Source domain, publication date, URL — used for country attribution via domain mapping

3.2 Analytical Approach

Public safety incidents were identified from 1,362 AIID incidents using keyword classification across six domain categories: law enforcement and surveillance, autonomous vehicles and transport, healthcare and medical, defense and national security, cybersecurity and critical infrastructure, and border and immigration control. This procedure identified 463 public safety incidents (34.0% of the database).

Country attribution was performed via two complementary methods: (1) AIID deployer country field, which provided attribution for approximately 37% of incidents; and (2) source domain mapping of 7,967 media reports to reporting country through a manually constructed domain-to-country lookup table covering major media outlets in 15 countries. Combined, these methods produced country attribution for 700 media reports linked to public safety incidents.

Population-normalized incident rates (per 10 million population) were computed for 40 countries with sufficient data. Pearson correlation coefficients were computed between governance score and incident rates. Statistical significance was tested at $p < 0.05$. Governance tiers were defined as: Developing (0–55), Emerging (55–70), Advanced (70–80), Leading (80+).

3.3 Limitations

Three methodological limitations bear on interpretation of the empirical findings and are central to the paper's theoretical argument:

- Country attribution gap: 63% of public safety incidents have no attributed country of deployment. This limitation is itself a primary empirical finding — it reflects the current state of global AI incident monitoring infrastructure.
- English-language bias: The AIID is weighted toward English-language media ecosystems. Major AI deploying countries including China, Japan, South Korea, and Germany are systematically underrepresented, which biases cross-national comparisons.
- Causality: All reported correlations are observational. No causal claim is made about the direction of the governance-incident relationship. Longitudinal panel data tracking governance changes against subsequent incident rates would be required for causal inference.

4. EMPIRICAL FINDINGS

4.1 The Governance-Incident Correlation

Across 40 countries, the governance score from the Oxford Insights GARI 2024 correlates positively with public safety AI incidents per capita (Pearson $r = 0.363$, $p = 0.021$). This positive direction is counterintuitive from the perspective of the governance-centric paradigm, which would predict that higher governance scores should be associated with fewer incidents.

Governance Tier	Score Range	Avg PS Incidents per 10M	Ratio vs Developing
Developing	0 – 55	0.27	1.0× (baseline)
Emerging	55 – 70	0.58	2.1×
Advanced	70 – 80	1.87	6.9×
Leading	80+	3.00	11.1×

The 11-fold difference in documented incidents between Leading and Developing governance tier countries is inconsistent with governance causing harm reduction. Countries with the most mature governance frameworks are documenting substantially more incidents, not fewer.

4.2 The Reporting Infrastructure Interpretation

The positive governance-incident correlation is explicable through what we term the reporting infrastructure interpretation: high-governance countries possess the regulatory reporting requirements, media transparency norms, press freedom, and accountability ecosystems that make AI harms visible. Lower-governance countries likely experience similar or greater harms in absolute frequency, but those harms go undocumented because the infrastructure to surface them does not exist.

This interpretation is supported by analysis of media coverage depth across governance tiers. Incidents in Advanced and Leading tier countries generate significantly more media reports per incident (median 3–4 reports per incident) compared to Developing tier countries (median 1–2). High-governance countries do not merely document more incidents; they document each incident more thoroughly, suggesting an active and competitive accountability ecosystem rather than simple regulatory compliance.

4.3 The UK Outlier

The United Kingdom presents the most analytically significant case in the dataset. Despite a lower composite governance score than the United States (78.88 vs. 87.03), the UK

generates 12.2 documented public safety AI incidents per 10 million population, compared to 7.5 for the United States. This divergence cannot be explained by the composite governance index alone.

The UK's elevated incident documentation rate is consistent with a distinct accountability culture: aggressive AI accountability journalism concentrated in outlets such as The Guardian, the BBC, and The Register; more stringent regulatory posture on facial recognition and surveillance AI than the equivalent US position; and active enforcement activity by the Information Commissioner's Office. The UK case demonstrates that governance quality — the behavioral culture of accountability — and governance score — the composite index ranking — are not identical. Accountability culture may operate as an independent behavioral constraint beyond what formal governance frameworks capture.

4.4 Sectoral Concentration and Growth Trajectory

Of 463 public safety incidents, 190 (41.0%) involve law enforcement and surveillance applications, representing a larger share than autonomous vehicles (77, 16.6%), healthcare (49, 10.6%), border and immigration control (28, 6.1%), and defense and national security (25, 5.4%) combined. This concentration has significant implications for governance design: general-purpose AI governance frameworks may be less effective than sector-specific professional accountability structures with domain-specific ethics cultures.

Documented public safety incidents have grown from 35 in 2022 to 79 in 2024, a 126% increase over two years. This growth trajectory indicates that AI deployment in high-stakes public safety domains is accelerating substantially faster than governance frameworks are adapting. The behavioral sufficiency gap, if it exists, is widening.

5. THE BEHAVIORAL SUFFICIENCY PROBLEM: A THEORETICAL FRAMEWORK

5.1 Two Lenses on the AI-Human Analogy

The Behavioral Sufficiency Problem can be approached through two distinct analytical lenses, corresponding to two positions on the relevant similarity between AI systems and human agents. These lenses lead to different prescriptive recommendations, but converge on the same diagnosis: governance alone is insufficient. We present them in parallel because the argument does not depend on resolving the contested empirical question of AI-human similarity.

Lens A — The Parity Argument	Lens B — The Asymmetry Argument
Core claim: AI systems now approximate human reasoning closely enough that they require the same behavioral scaffolding humans require — not just external rules, but internalized values and cultural constraints.	Core claim: AI systems are sufficiently different from humans that governance is even less sufficient — because AI lacks all the non-governance behavioral constraints that humans have by default. Governance is doing work it was never designed to do alone.
Key evidence: — Chain-of-thought reasoning approximates deliberation — Context sensitivity resembles situational judgment — Output quality rivals expert human performance across domains	Key evidence: — No peer community or professional identity — No reputation accumulation across contexts — No intrinsic motivation or social shame mechanisms
Prescriptive implication: Solutions live inside the AI system — constitutional training, value alignment, identity-level behavioral constraints embedded in the model itself.	Prescriptive implication: Solutions live in the human institutions around the AI — organizational ethics culture, deployment community norms, sector-specific professional accountability structures.
Point of Convergence: Regardless of which lens is correct, the conclusion is identical — governance frameworks alone are not behaviorally sufficient for AI systems operating in high-stakes domains.	

5.2 What Fills the Gap in Human Systems

Understanding what governance does not do for human behavior requires mapping the non-governance behavioral constraints that actually do most of the work. We identify four primary mechanisms:

Professional Ethics Culture

Occupational communities establish behavioral norms that are enforced through professional socialization, certification, peer review, and community sanction rather than primarily through law. A physician who conceals a medical error faces professional community consequences — loss of respect, exclusion, informal sanction — that operate independently of and often more powerfully than legal liability. AI systems have no analogous professional community.

Accountability Culture and Peer Judgment

Human behavior is strongly influenced by the anticipated judgments of peers whose opinions are valued. The prospect of professional disapproval from respected colleagues is a powerful behavioral constraint. This mechanism requires a stable peer community with shared values and ongoing interaction — conditions that AI systems do not currently meet.

Internalized Identity and Moral Self-Concept

Humans develop stable self-concepts that include moral commitments. Behavior inconsistent with self-concept produces psychological discomfort that functions as an intrinsic behavioral constraint. This mechanism does not require external observation or sanction to operate — it functions even when no one is watching. Current AI systems optimize for objectives rather than values; they have no analogous intrinsic motivational structure.

Social Feedback Loops and Real-Time Correction

Human behavior is continuously calibrated by social feedback — responses from community members that signal norm deviation. These feedback loops provide real-time correction that does not require formal regulatory intervention. AI systems in deployment receive feedback through model updates and policy changes, but these mechanisms operate on much longer timescales and require human intermediaries.

5.3 The Implications for AI Governance Design

The Behavioral Sufficiency Problem does not imply that AI governance frameworks are useless. Governance serves critical functions: it establishes minimum floors for behavior, creates accountability mechanisms, mandates transparency, and provides recourse when harms occur. These are necessary contributions.

The problem is that governance frameworks are being asked to serve as sufficient behavioral constraint for systems that approximate human reasoning without possessing the behavioral infrastructure that makes governance work for humans. The result is a set of conditions under which:

- Governance compliance may not predict behavioral safety, because governance is designed to constrain agents with internal behavioral orientation, not agents without any.
- Incident databases will systematically underrepresent actual harm, because harm visibility is itself a function of governance infrastructure, as the empirical findings demonstrate.
- Evidence-based governance improvement becomes impossible, because the feedback loops that would allow governance to adapt to observed outcomes are incomplete.

The governance gap is not a gap in the volume of frameworks, principles, or regulations. It is a gap in the behavioral infrastructure that governance was designed to complement. Filling this gap requires understanding what the analogs to professional culture, peer accountability, and moral identity look like for AI systems and the organizations that deploy them.

6. RECOMMENDATIONS

6.1 For Policymakers

- **Priority:** Establish mandatory AI incident reporting requirements as a prerequisite for AI deployment in high-stakes public safety domains. The empirical findings demonstrate that incident documentation is a function of governance infrastructure, not incident frequency. Mandatory reporting is the behavioral equivalent of professional accountability — it creates the visibility without which governance cannot adapt.
- Invest in multilingual, multi-jurisdictional incident monitoring infrastructure. The 63% attribution gap in current incident databases represents a fundamental constraint on evidence-based AI governance. The OECD AI Incidents Monitor and AIID expansion efforts require public funding commensurate with this priority.
- Design sector-specific governance frameworks that leverage existing professional ethics cultures rather than replacing them with generic AI regulation. Healthcare AI should be governed within medical ethics frameworks. Law enforcement AI should be governed within police accountability structures. Professional culture provides the behavioral scaffolding that generic governance frameworks cannot replicate.

6.2 For Deploying Organizations

- **Strategic priority:** Treat organizational AI ethics culture as a governance mechanism in its own right. The UK evidence suggests that accountability culture — the informal norms, peer expectations, and professional identity of the people deploying AI systems — may be as important as formal governance compliance. Organizations should invest in building AI accountability cultures rather than only implementing governance frameworks.
- Implement internal incident reporting and near-miss documentation systems before external regulatory requirements mandate them. Internal visibility of AI system behavior is the organizational analog to the social feedback loops that constrain human behavior.
- Recognize that AI governance compliance and AI behavioral safety are not equivalent. Governance frameworks establish minimum floors; behavioral safety in high-stakes domains requires the full range of organizational, cultural, and technical mechanisms described in this paper.

6.3 For AI Developers

- **Technical direction:** Invest in constitutional training approaches that embed behavioral constraints at the model level rather than relying exclusively on external governance compliance. Constitutional AI methods represent the closest current analog to internalized values — the mechanism by which human behavior is constrained even when external observation is absent.
- Develop evaluation benchmarks specifically for behavioral robustness under governance-absent conditions — scenarios in which external oversight is reduced or ambiguous. Current safety evaluations typically assess behavior under conditions of explicit

prompting; real-world deployment frequently involves contexts where governance expectations are unclear.

7. CONCLUSION

The Behavioral Sufficiency Problem is not an argument against AI governance. It is an argument that the AI policy community has systematically underspecified what sufficient conditions for safe AI behavior actually require.

Human regulatory theory was developed to govern agents embedded in rich social, cultural, and psychological ecosystems. Those ecosystems do most of the behavioral work; governance is the backstop. When governance frameworks are imported into AI contexts where those ecosystems do not exist, the frameworks are being asked to carry a load they were not designed to bear.

The empirical evidence in this paper — a positive, statistically significant correlation between governance maturity and documented AI incidents — is consistent with this theoretical claim. It suggests that what governance primarily produces is visibility, not safety. Visibility is valuable and necessary, but it is not sufficient.

The agenda implied by this analysis is substantial. It requires developing the AI equivalents of professional culture, peer accountability, and internalized values — whether inside AI systems through constitutional training methods, or in the human institutions and communities around those systems through organizational ethics culture, mandatory reporting infrastructure, and sector-specific professional norms.

This is harder than writing governance frameworks. It may be more important.

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